Data

month	branch_id	city	product_type	active_customers	total_balance_usd	monthly_transactions
2025-01	BR001	New York	Savings Account	420	2604000	420
2025-01	BR002	Boston	Checking Account	510	1428000	680
2025-01	BR003	Chicago	Credit Card	350	630000	920
2025-01	BR004	Dallas	Home Loan	125	23125000	310
2025-01	BR005	Seattle	Business Loan	95	9025000	240
2025-02	BR001	New York	Checking Account	435	1218000	460
2025-02	BR002	Boston	Savings Account	390	2418000	405
2025-02	BR003	Chicago	Credit Card	370	666000	980
2025-02	BR004	Dallas	Home Loan	130	24050000	325
2025-02	BR005	Seattle	Business Loan	102	9690000	255
2025-03	BR001	New York	Credit Card	365	657000	950
2025-03	BR002	Boston	Home Loan	118	21830000	295
2025-03	BR003	Chicago	Savings Account	445	2759000	430
2025-03	BR004	Dallas	Business Loan	110	10450000	280
2025-03	BR005	Seattle	Checking Account	520	1456000	700

Objective

To analyse the above dataset using various tools such as numpy, pandas etc..

Steps

- **Step 1:** Downloading dataset
- **Step 2:** Install python
- **Step 3:** create virtual environment
- **Step 4:** Install and import the required libraries(numpy, pandas)
- **Step 5:** Load the dataset using pandas
- **Step 6:** Use the library functions to explore and analyse the above dataset to get the following outcomes
    - What is the average number of monthly transactions?
      ```
      df["monthly_transactions"].mean()
      ```
    - What is the total balance across all records?
      ```
      df["total_balance_usd"].sum()
      ```
    - What are the unique cities in the dataset?
      ```
      df["city"].unique()
      ```
    - What are the cities with the highest total active customers?
      ```
      df.groupby("city")["active_customers"].sum().sort_values(ascending=False
      )
      ```
    - What is the average number of active customers for each product type?
      ```
      df.groupby("product_type")["active_customers"].mean()
      ```